r/NoStupidQuestions

How are AI servers using all the water?

Posted by u/Medium-Sized-Jaque • • 0 points • 9 comments

I keep hearing about how AI is using all the water to cool the servers. I understand that needs a lot of cooling but what I don't understand is why it needs continuous water. With a desktop water cooler the system is closed, the hot water goes to the radiator the fans cool it off and it cycles back through. Do they not do the same on server farms? Do they just let the water evaporate into the air instead of reusing it? Is it actually a closed system but the small imperfections add up at that scale?

Comments

u/[deleted] • 1 points • 2026-03-16 18:42 UTC

[removed]

u/jokkaM • 2 points • 2026-02-07 07:14 UTC

vale mira así es pero piensa que todos hacen eso la IA tik tok Google Microsoft y mil empresas más, no es algo nuevo

> **u/Medium-Sized-Jaque** • 1 points • 2026-02-07 07:19 UTC
>
> I assumed it wasn't a new thing. Just that the number of AI data centers being built brought it to public attention. I was more curious about the why they lose water instead of using a heat exchanger.

u/Delehal • 1 points • 2025-09-17 20:23 UTC

>Do they just let the water evaporate into the air instead of reusing it?

Yes, this. A lot of data centers rely on evaporative cooling.

> **u/Medium-Sized-Jaque** • 2 points • 2025-09-17 21:03 UTC
>
> Very simple and straightforward. Thank you.

u/ion_driver • 0 points • 2025-09-17 20:07 UTC

Waste heat gets dumped into the environment, same as with power plants. Either water us evaporated in a cooling tower or water is taken from a body of water such as a river or bay and returned a few degrees hotter. If a data center were to use HVAC to cool it, then the power provided by a power plant would involve an amount of waste heat being dumped to the environment.

Its partly a nothing-burger and partly a misunderstanding of how energy is produced and used. Its more efficient to evaporate water to cool a data center than it would be to use HVAC.

If data centers were placed in an area where power could be powered by solar/wind/hydro, then they could maybe cool down without using so much water. The problem is that these energy sources are not reliable enough to power something that uses a large amount of power at all times.

u/SelleckssLastStand • 1 points • 2025-09-17 20:05 UTC

I think it just has to do with scale.

A small closed-loop system is fine for a PC - it's a small amount of liquid that is just pumped around your case. But when you are talking about gigantic server farms - that's a huge amount of water you have to push around. Not only is evaporation more efficient at cooling, but it's a less complex process to engineer.

u/Penis-Dance • 1 points • 2025-09-17 20:05 UTC

They use the water to cool the equipment which causes the water to evaporate into the atmosphere.

u/Astramancer_ • 2 points • 2025-09-17 20:04 UTC

It's because they need a *lot* of cooling. Your water-cooled computer is actually air-cooled. It uses water to move heat from the computer to the radiator and then air-cools the water in the radiator.

That works great, until you're generating more heat than you can reasonably get rid of using air cooling. Which server farms do. So they dump a large amount of hot water down the drain or evaporated into the air.

There are uses for that hot water, but not every installation will be located in reasonable proximity to the kinds the things that need that hot water.

> **u/Medium-Sized-Jaque** • 1 points • 2025-09-17 21:05 UTC
>
> So basically the radiator(s) needed for such an amount wouldn't be practical?
>
> > **u/Astramancer_** • 1 points • 2025-09-17 21:06 UTC
> >
> > Radiators and associated infrastructure, yet.

u/crashorbit • 5 points • 2025-09-17 20:04 UTC

In large data centers the hot side of the AC system often includes evaporation cooling.